QUESTION 1

a. Describing the crabs dataset

```
library("MASS")

?crabs
```

Description

The crabs data frame has 200 rows and 8 columns, describing 5 morphological measurements on 50 crabs each of two colour forms and both sexes, of the species *Leptograpsus variegatus* collected at Fremantle, W. Australia.

b. loading and displaying the structure

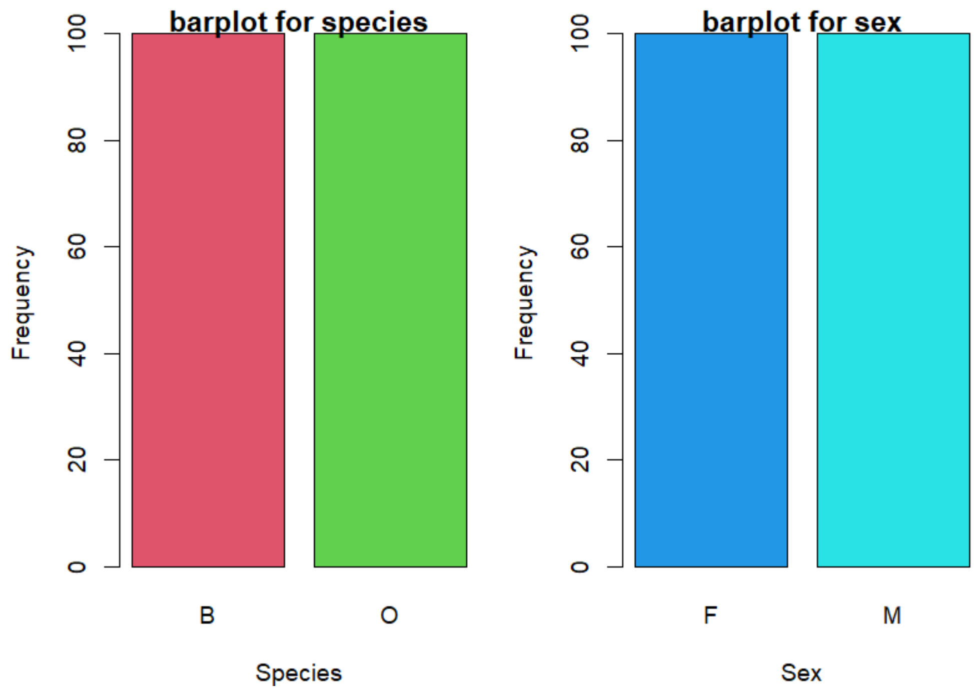
```
data(crabs) ## To laod the dataset

head(crabs) # to have a preview of the data


str(crabs) # For checking the structure of the dataset
```

c. a frequency table for species and sex

```
?table

f_tab_species <- table(crabs$sp)

f_tab_species


f_tab_sex <- table(crabs$sex)
```

bar chart summarizing the groups composition

```
barplot(f_tab_species, main = "barplot for species",
        xlab = "Species",
        ylab = "Frequency",
        col = 2:3)
barplot(f_tab_sex, main = "barplot for sex",
        xlab = "Sex",
        ylab = "Frequency",
        col = 4:5)
```

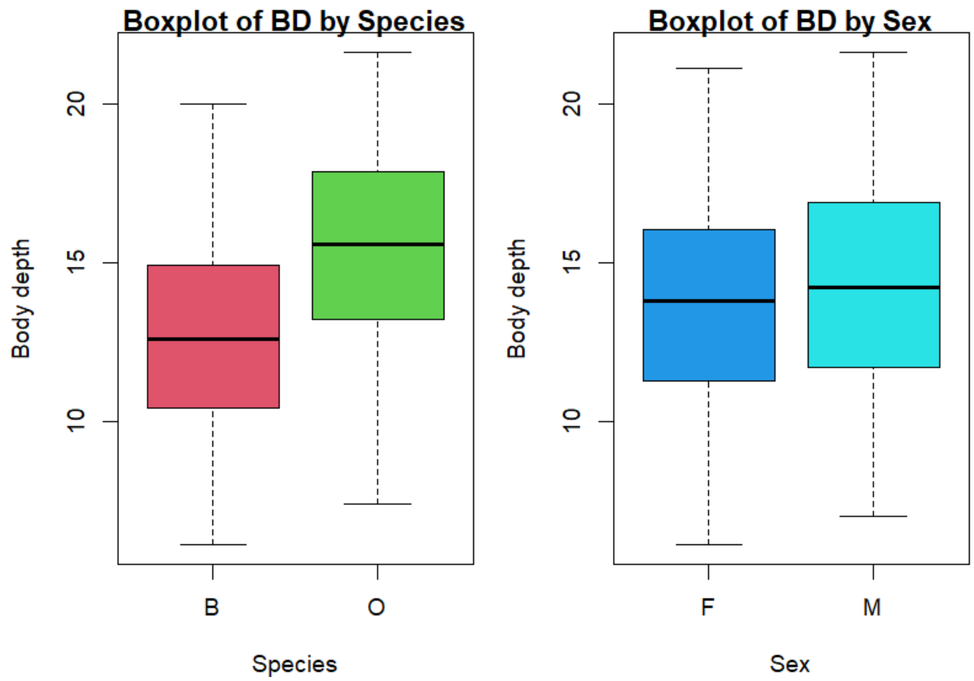


e. Across all four groups the dataset is balanced

QUESTION 2

```
par(mfrow = c(1, 2), mar = c(4,4,1,1))
boxplot(crabs$BD ~ crabs$sp,
    main = "Boxplot of BD by Species",
    xlab = "Species",
    ylab = "Body depth",
    col = 2:3)
boxplot(crabs$BD ~ crabs$sex,
    main = "Boxplot of BD by Sex",
    xlab = "Sex",
    ylab = "Body depth",
    col = 4:5)
```

a.

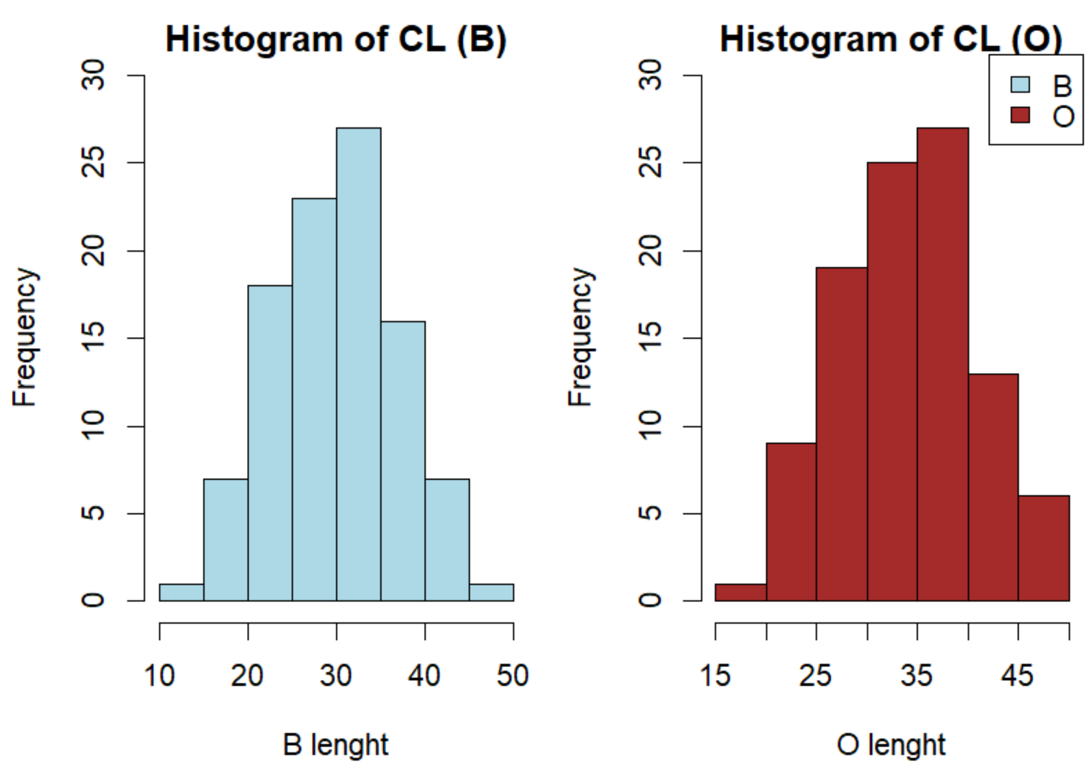


b. from the boxplots there are no outliers .

the species O has the highest median of body depth and species B has the lowest median of body depth.

QUESTION 3

```
hist(crabs$CL[crabs$sp == "B"],
    main = "Histogram of CL (B)",
    xlab = "B lenght", col = "lightblue",
    ylim = c(0,30))
hist(crabs$CL[crabs$sp == "O"],
    main = "Histogram of CL (O)",
    xlab = "O lenght", col = "brown",
    ylim = c(0,30))


legend("topright", legend = c("B", "O"),
     fill = c("lightblue", "brown"))
```



```
library(psych)


describe(crabs$CL[crabs$sp == "B"])

describe(crabs$CL[crabs$sp == "O"])
```

b.

```
> library(psych)
> describe(crabs$CL[crabs$sp == "B"])
   vars   n  mean  sd median trimmed  mad  min  max range skew kurtosis   se
X1    1 100 30.06 6.9   30.1   30.02 7.49 14.7 47.1  32.4 0.03    -0.65 0.69
> describe(crabs$CL[crabs$sp == "O"])
   vars   n  mean   sd median trimmed  mad  min  max range  skew kurtosis   se
X1    1 100 34.15 6.76  34.55   34.29 7.56 16.7 47.6  30.9 -0.15    -0.56 0.68
```

From the describe function and the histograms, the shapes of both are approximately normal

The species has skewness 0.03 and kurtosis -0.65 and the sex has skewness -0.15 and kurtosis -0.56

The mode of the species is 47.1 and sex is 47.6

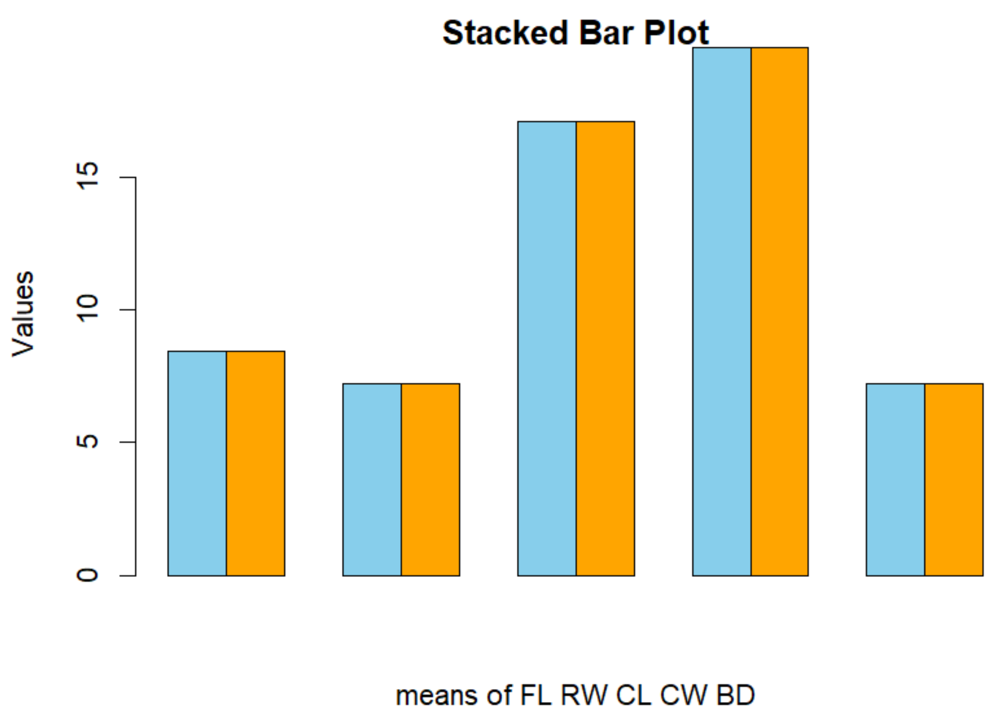
QUESTION4

```r
mean_FL_sp <- mean(crabs$FL[crabs$sp])

mean_RW_sp  <- mean(crabs$RW[crabs$sp])

mean_CL_sp  <- mean(crabs$CL[crabs$sp])

mean_CW_sp  <- mean(crabs$CW[crabs$sp])

mean_BD_sp  <- mean(crabs$BD[crabs$sp])


mean_FL_sex <- mean(crabs$FL[crabs$sex])

mean_RW_sex  <- mean(crabs$RW[crabs$sex])

mean_CL_sex  <- mean(crabs$CL[crabs$sex])

mean_CW_sex  <- mean(crabs$CW[crabs$sex])

mean_BD_sex  <- mean(crabs$BD[crabs$sex])


all_means <- data.frame(

  measurement = c("mean_FL", "mean_RW", "mean_CL", "mean_CW","mean_BD"),

  Species_Mean  = c(mean_FL_sp, mean_RW_sp, mean_CL_sp, mean_CW_sp,
mean_BD_sp),

  Sex_Mean  = c(mean_FL_sex, mean_RW_sex, mean_CL_sex, mean_CW_sex,
mean_BD_sex)

)


print(all_means)
```

```
> print(all_means)
  measurement Species_Mean Sex_Mean
1     mean_FL         8.45     8.45
2     mean_RW         7.20     7.20
3     mean_CL        17.10    17.10
4     mean_CW        19.90    19.90
5     mean_BD         7.20     7.20
```

b. The grouped Barchat

```
Species_Mean  <-  c(mean_FL_sp, mean_RW_sp, mean_CL_sp, mean_CW_sp,
mean_BD_sp)

Sex_Mean  <- c(mean_FL_sex, mean_RW_sex, mean_CL_sex, mean_CW_sex,
mean_BD_sex)


mean_matrix <- rbind(Species_Mean,Sex_Mean)

par(mfrow = c(1, 1))

barplot(mean_matrix,

      col = c("skyblue", "orange"),

      main = "Stacked Bar Plot",beside = T,

      xlab = "means of FL RW CL CW BD",
```



```
?pairs

pairs(mean_matrix)
```